

Optimizing Secrecy Rate for Federated Learning Model Aggregation With Intelligent Reflecting Surface Toward 6G Ubiquitous Intelligence

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Abstract—Non-Orthogonal Multiple Access (NOMA) based Federated Learning (FL) can achieve the massive connectivity of Internet of Thing (IoT) devices, high transmission rate, and pervasive intelligence in 6G networks. However, the stochastic channels and frequent model parameter updates may incur degraded transmission rate and diminished FL performance, while privacy leakage may happen if Eavesdroppers (Eves) intercept the FL training process. To address the above issues, we exploit Intelligent Reflecting Surface (IRS) to reconfigure wireless signal propagation for secure transmission and fast convergence of NOMA-based FL. In this article, a Deep Reinforcement Learning (DRL) based approach is proposed to jointly optimize the transmission power of edge devices and IRS phase shift to maximize the minimum secrecy rate in the model parameter uploading process. Numerical results validate the efficiency of our proposed algorithm and demonstrate that IRS can improve the secrecy rate.

Index Terms—Federated learning (FL), non-orthogonal multiple access (NOMA), intelligent reflecting surface (IRS), secrecy rate.

I. INTRODUCTION

6G IS envisioned to enable diverse Internet of Things (IoT) device accessibility coupled with ultra-low end-to-end delay, while also promising ubiquitous intelligence. Federated Learning (FL), as a decentralized model training method, plays

a pivotal role in realizing the above key functionalities. FL aims to train a high-quality Machine Learning (ML) model via the cooperation of multiple edge devices [1]. It can facilitate the customization of AI services and cater to individual preferences, contributing to the user-centric 6G experience. Moreover, FL empowers real-time learning on edge devices to reduce the back-and-forth communications with the centralized server, which is crucial for future latency-sensitive 6G applications including autonomous driving, metaverse, and smart city. Additionally, as 6G shifts the concept from data-centric, machine-centric, and application-centric towards human-centric [2], the challenge of data privacy preserving Artificial Intelligence (AI) can be alleviated by keeping data localized for training [3], [4].

Due to the uneven user distribution and data heterogeneity in 6G networks, it is necessary to involve a large number of devices to build a reliable learning model swiftly [5]. Nevertheless, the constraints posed by limited resources imply that only a few users can be allowed to participate in collaborative FL training [6], resulting in a decreased accuracy rate. To address the resource constraints, Non-Orthogonal Multiple Access (NOMA) has been studied to integrate with FL [7] since it can enhance frequency efficiency via allowing a large number of wireless devices to share radio blocks concurrently [5]. However, the stochastic wireless channels of NOMA-based FL and frequent communication between Base Station (BS) and edge devices result in an unbearable degraded transmission rate and diminished FL training performance. Moreover, the complex propagation of NOMA-based FL can lead to security concerns [8], [9], [10] as shown in Fig. 1.

To tackle the above concerns, Wang et al. [8] and Qian et al. [9] adopt a model spread method using chaotic sequences to safeguard the trained model delivered through NOMA from malicious nodes. Zheng et al. [10] discuss security concerns in an Over-The-Air (OTA) communication-based FL system, which shares the same wireless channel characteristics as the NOMA-based FL system. They utilize an Intelligent Reflecting Surface (IRS) to enable covert transmission by configuring the maximum training participants and IRS phase shift. IRS has been an emerging advanced technology that can compensate for degraded rate and privacy leakage which can reconfigure the wireless propagation environment to improve spectrum and energy efficiency [11], [12]. In [13], integrating IRS with a near-space information network

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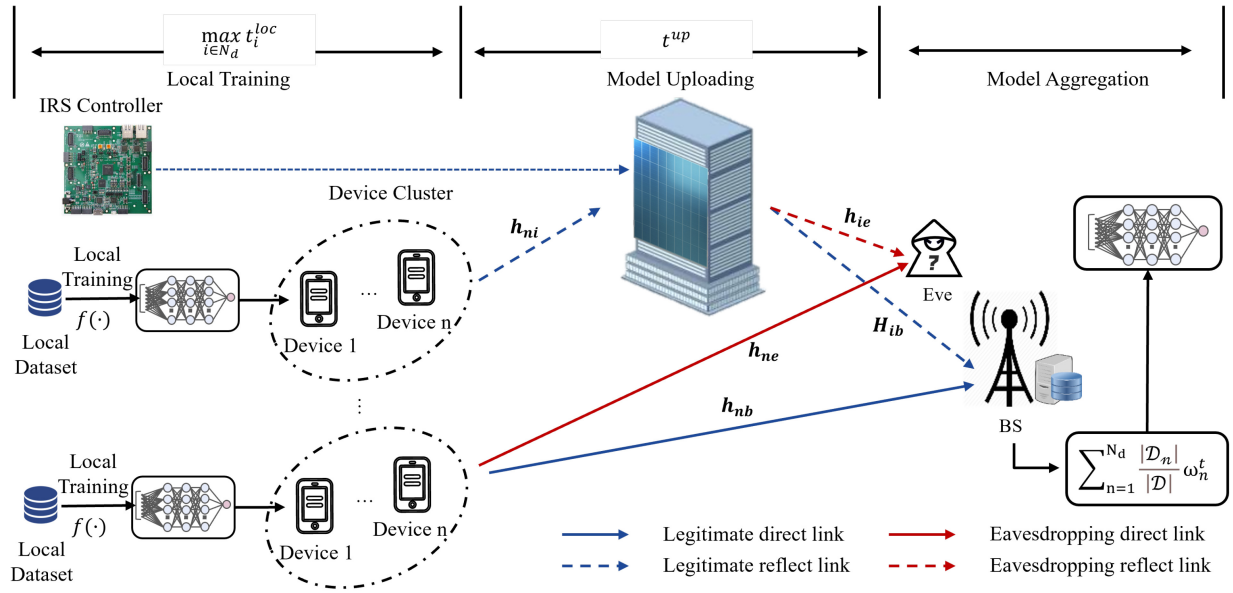


Fig. 1. IRS-assisted NOMA-based FL system with the existence of Eve.

significantly enhances energy efficiency. With the aid of the IRS, the effective throughput is maximized, and energy consumption is minimized greatly. Since IRS mainly consists of passive reflecting elements [14], it proactively adjusts the wireless signal at a lower cost and in a more efficient full-duplex mode compared to other traditional relay methods [15], [16]. With the IRS controller, each element can tune the amplitude and phase shift independently, thereby collaboratively changing the reflected signal propagation [15]. In this way, the IRS can enhance the channel gain from information sources to legitimate users and suppress the signal to Eavesdroppers (Eves), thereby enhancing the transmission rate and security. In [17], multiple IRSs have been deployed in a millimeter-wave system to improve the secrecy rate.

Traditional security methods primarily concentrate on the addition of jamming signals. End devices [18] and Unmanned Aerial Vehicles (UAVs) [19] have been studied to serve as jammers to impede the Eves. However, the addition of jamming signals not only results in additional energy consumption for resource-constrained IoT devices, but also significantly degrades FL performance. Although [10] tackles the covert-ness problem in OTA communication-based FL, the OTA performs inflexibly in supporting diverse service requirements during FL aggregation. To address these issues, we explore the benefits of the IRS to boost the security and convergence speed of NOMA-based FL, enabling the pervasive intelligence and massive connectivity of the upcoming 6G. Our key contributions include:

- We integrate IRS with NOMA-based FL to tackle security concerns. IRS is expected to improve the device-BS channel while impeding the wire-tap channel from the device to Eve.
- We define the secrecy rate as the difference between the rates from devices to BS and Eve. We then formulate a minimum secrecy rate maximization problem subjecting to the transmission power budget and IRS phase shift.

- To solve the above problem, the Deep Deterministic Policy Gradient (DDPG) based approach is proposed to jointly optimize the transmission power and IRS phase shift.

The remainder of this manuscript is organized as follows. In Section II, we provide a comprehensive review of the existing literature. Then, in Section III, we present the system model of the IRS-assisted NOMA-based FL and formulate the objective function. Following that, the DDPG algorithm for jointly designing the transmission power and IRS phase shift is introduced to address the optimization issue in Section IV. Then we perform several simulations to validate the effectiveness of our proposed solution and present the results to verify our proposal in Section V. Finally, we conclude our manuscript in Section VI.

Notations: We use the bold uppercase and lowercase letters to denote matrices and vectors, respectively. $\text{diag}(\mathbf{x})$ means the diagonal matrix of $\mathbf{x}$, $|x|$ denotes the absolute value of the scalar x . $\Re(x)$ and $\Im(x)$ represent the real part and imaginary part of the complex value x , respectively. $\mathbb{C}^{X \times Y}$ denotes the complex matrices with size $X \times Y$.

II. RELATED WORKS

In this section, we investigate the research in three aspects: FL privacy protection, NOMA-based FL, and IRS-assisted NOMA-based FL.

(1) *FL privacy protection:* Differential Privacy (DP), compression, and encryption are usually used to secure the shared gradients. Reference [20] puts forward a new two-layer DP-based gradient-protected approach. Users' training data are first compressed utilizing Hense's Lemma, following that data privacy is realized through the addition of noise to further perturb the compressed data in the second layer. A dual approach utilizing both Local DP (LDP) and Central DP (CDP) is employed in another study [21]. The authors utilize

the Gaussian Mechanism to achieve (ϵ, δ) -LDP, ensuring privacy in the contributions made by devices. Differently, [22] and [23] choose to encrypt the gradients. The authors in [22] propose a noninteractive privacy-enhanced FL scheme where the gradients are perturbed and encrypted prior to transmission. Traditional homomorphic encryption where all devices share the same encryption and decryption keys can lead to gradient disclosure if any participant colludes with the adversary. To avoid this, [23] adopts Paillier homomorphic encryption and random number generation technique to secure the gradients.

(2) *NOMA-based FL*: NOMA is usually adopted into the FL system to utilize the superposition property of wireless channels and improve spectrum efficiency. It enables the participation of multiple devices in FL training to improve performance. Reference [7] utilizes the NOMA to meet the stringent latency requirements. In this work, the authors propose a fair resource-sharing scheme to minimize the total latency of FL considering the power and energy constraint. To further speed up the FL training, the uploading parameters are compressed in some work [24], [25]. The state-of-the-art wireless power transfer (WPT) technique is also introduced into the NOMA-based FL system [26], [27]. In [26], the utilization of NOMA leads to high energy consumption, resulting in extended WPT duration and consequently degrading the FL performance. The authors try to balance the trade-off between energy consumption and FL training performance. Similarly, in [27], energy consumption is reduced via jointly optimizing several communication and computation parameters.

(3) *IRS-assisted NOMA-based FL*: IRS has been utilized to alleviate the drawbacks of OTA or NOMA channels [6], [28]. OTA can enable the one-shot model aggregation of FL, while FL performance is often compromised by the inherent noise and the consequent reduced transmit power for signal alignment. IRS has been adopted then to address this issue. In [5], IRS has been deployed to improve the weak channel, collaborating with NOMA to speed up the FL convergence. A long-term design scheme is proposed to assign the bandwidth and other resources to specific training rounds, improving the prediction accuracy [28]. IRS has also been deployed to tackle the hybrid user problem, where communication-centric and learning-centric users are considered to enable massive connectivity and ubiquitous intelligence. It is introduced to coordinate the decoding order of these user types, enhancing interference mitigation, and extending coverage [29]. Privacy concerns are also considered in the deployment of IRS. Reference [10] improves the physical layer security via deploying IRS, where the optimal covertness level is reached by enabling zero force.

Extensive research has been conducted on the IRS-assisted uplink NOMA for FL. However, it remains an open question regarding the security in this scenario. As security and privacy issues are of great importance in future 6G networks, we leverage the IRS to reconfigure the wireless signal propagation and secure the NOMA-based FL.

III. SYSTEM MODEL AND PROBLEM FORMULATION

We consider an IRS-assisted NOMA-based FL system in Multi-User Multiple-Input Multiple-Output as shown in Fig. 1,

including one BS with N_T antennas, N_d devices with a single antenna, an Eve with a single antenna, and one IRS with N_I passive reflecting elements. The Eve is situated in the close proximity to the BS and intercepts the parameters from devices to BS. The sets of devices and IRS reflecting elements can be denoted by $\mathcal{N}_d = \{1, 2, \dots, N_d\}$ and $\mathcal{N}_I = \{1, 2, \dots, N_I\}$. The channel state information is assumed available for all nodes including IRS [30]. Additionally, we assume that the channel coefficients are quasi-static [31] so that they can be regarded as constant during FL iterations.

A. FL Model

For the FL, we consider the Federated Averaging (FedAvg) algorithm. We assume that N_d devices participate in the local training. The device n collects and holds its data set of $\mathcal{D}_n = \{(\mathbf{x}_i, y_i)\}_{i=1}^{|\mathcal{D}_n|}$, where $(\mathbf{x}_i, y_i)$ is the i -th data pair of the device n consisting of the input $\mathbf{x}_i$ and its label y_i , and $|\mathcal{D}_n|$ denotes the cardinality of the data set $\mathcal{D}_n$.

The local training process of device n is performed via minimizing the local loss function:

$$\min_{\omega} F_n(\omega) = \frac{1}{|\mathcal{D}_n|} \sum_{j=1}^{|\mathcal{D}_n|} f(\mathbf{x}_j, y_j; \omega_j), \quad (1)$$

where $f(\mathbf{x}_j, y_j; \omega_j)$ is the loss of prediction on the local data pair $(\mathbf{x}_j, y_j)$ calculated with the model parameters ω_j .

Following that, the BS performs the global training in a way that minimizes the global loss function:

$$\min_{\omega} f(\omega) = \sum_{n=1}^{N_d} \frac{|\mathcal{D}_n|}{|\mathcal{D}|} F_n(\omega), \quad (2)$$

where $|\mathcal{D}| = \sum_{n=1}^{N_d} |\mathcal{D}_n|$. In the FL training process, the BS first initializes the model parameter ω^0 . Subsequently, during local training and global aggregation, the Stochastic Gradient Descent (SGD) method is executed to update the parameters. Following is the general process of FL in training round t :

(1) *Model distribution*: BS distributes the present global model ω^t to the participated devices.

(2) *Local training*: Local devices train their respective models with the received global model using their own data and update the gradient $\mathbf{g}_n = \nabla F_n(\omega)$. Local model is updated via executing the computation $\omega_n^t = \omega_n^{t-1} - \eta \mathbf{g}_n$, where η is the learning rate for parameter update. In each training round, the computation can be executed $\frac{|\mathcal{D}_n|}{M}$ times in device n , where M is the batch size.

(3) *Model uploading*: Participated local devices upload the updated model ω_n^t to the BS.

(4) *Model aggregation*: BS aggregates the global model based on the received ω_n^t using a weighted average method: $\omega^{t+1} = \sum_{n=1}^{N_d} \frac{|\mathcal{D}_n|}{|\mathcal{D}|} \omega_n^t$.

B. Communication Model

We assume that the channel experiences both large-scale path loss and small-scale fading. The path loss follows the log-distance path loss model which considers the environmental conditions. It can be expressed as $\text{PL} = \text{PL}_0 - 10\zeta \log_{10} \frac{d}{d_0}$, where PL_0 is the path loss with the unit of decibels (dB) at

the reference distance d_0 , ζ is the corresponding path loss exponent, which depends on the environment and frequency, and d is the distance between the transmitter and receiver. The small-scale fading adopts the Rayleigh fading model.

Let $\mathbf{h}_{nb} \in \mathbb{C}^{1 \times N_T}$, $\mathbf{h}_{ne} \in \mathbb{C}^{1 \times 1}$, and $\mathbf{h}_{ni} \in \mathbb{C}^{1 \times N_I}$ represent the channel gains from device n to BS, Eve, and IRS, respectively. The channels from IRS to the BS and Eve are denoted as $\mathbf{H}_{ib} \in \mathbb{C}^{N_I \times N_T}$ and $\mathbf{h}_{ie} \in \mathbb{C}^{N_I \times 1}$, respectively. The IRS has the phase shift $\Phi = \{e^{j\phi_1}, \dots, e^{j\phi_{N_I}}\}^T \in \mathbb{C}^{N_I \times 1}$, where $\phi \in [0, 2\pi]$. We assume the ideal reflection of IRS without the loss of signal power [32], i.e., each element $\Phi_l = e^{j\phi_l}$, $l \in N_I$ satisfies $|\Phi_l|^2 = 1$.

To avoid dealing with extremely small quantities and mitigate potential numerical issues, we make the normalization of the channel coefficients as $\mathbf{h}_{nb} \leftarrow \mathbf{h}_{nb}/\sigma$, $\mathbf{h}_{ne} \leftarrow \mathbf{h}_{ne}/\sigma$, $\mathbf{h}_{ni} \leftarrow \mathbf{h}_{ni}/\sigma$, and $\mathbf{h}_{ie} \leftarrow \mathbf{h}_{ie}/\sigma$, where σ^2 represents the power of the background white Gaussian noise. The effective channels between device n and BS, as well as that between device n and Eve can be expressed as $\mathbf{h}_{nb}^{\text{eff}} = \mathbf{h}_{in} \text{diag}(\Phi) \mathbf{H}_{ib} + \mathbf{h}_{nb}$ and $\mathbf{h}_{ne}^{\text{eff}} = \mathbf{h}_{in} \text{diag}(\Phi) \mathbf{h}_{ie} + \mathbf{h}_{ne}$, respectively.

We focus on the NOMA-based model uploading phase. In round t , the participating devices upload their models ω_n^t to the BS simultaneously. ω_n^t should be encapsulated to transmit symbols before transmitting to the BS via NOMA channels, which is expressed as s_n^t . The signals from users are intended to the BS which can be intercepted by the Eve. We assume that BS and Eve follow the same communication scheme. Thus, the signals received by BS and Eve are given by:

$$y_c = \mathbf{H}_c \mathbf{P} \mathbf{s} + \beta_c, c \in \{b, e\},$$

where $\mathbf{H}_{nb} = \{\mathbf{h}_{1b}^{\text{eff}}, \dots, \mathbf{h}_{N_d b}^{\text{eff}}\} \in \mathbb{C}^{N_d \times N_T}$ and $\mathbf{H}_{ne} = \{\mathbf{h}_{1e}^{\text{eff}}, \dots, \mathbf{h}_{N_d e}^{\text{eff}}\} \in \mathbb{C}^{N_d \times 1}$ are the device-BS and device-Eve channel matrix, respectively. $\mathbf{P}$ is the transmission power matrix at devices and constrained by $\mathbf{P} \leq P_{\text{Max}}$, where P_{Max} is the transmission power budget at devices. $\mathbf{s}$ is the transmit symbol with unit power at devices. β_b and β_e are the additive white Gaussian noise at BS and Eve, respectively.

The receiver adopts the successive interference cancellation to decode signals from devices to attenuate inter-user interference in the NOMA-based network. The receiver starts by decoding the signal from the channel with the highest gain [33]. Assuming that the channel gain from device 1 to N_d gradually increases, the signal from the n^{th} device is the $(N_d - n + 1)^{\text{th}}$ to be decoded. Consequently, a weaker channel's signal can be viewed as interference to a stronger channel's signal [33]. As a result, the transmission rates from device n to the BS, denoted as R_{nb} , and from device n to the Eve, denoted as R_{ne} , can be expressed as follows:

$$R_{nc} = \log_2 \left(1 + \frac{p_n |\mathbf{h}_{nc}^{\text{eff}}|^2}{\sum_{j=1}^{n-1} p_j |\mathbf{h}_{jc}^{\text{eff}}|^2 + 1} \right), c \in \{b, e\}, \quad (3)$$

where $p_n, \forall n \in \mathcal{N}_d$ is the n^{th} column vector of the matrix $\mathbf{P}$. Therefore, the secrecy rate received at BS can be described as

$$R_{sn} = R_{nb} - R_{ne}. \quad (4)$$

C. Problem Formulation

In the NOMA-based FL system, the BS aggregates the global model once all users have uploaded their models. Consequently, the overall FL performance is contingent on the slowest user. In this paper, we aim to maximize the minimum secrecy rate among users via jointly designing the transmission power at devices and IRS phase shift, while adhering to the transmission power constraint at users. Consequently, the optimization problem is formulated as:

$$\begin{aligned} & \text{maximize} \quad \min R_{sn} \\ & \text{s. t. : } \mathbf{P} \leq P_{\text{Max}}, \\ & \quad |\Phi_l|^2 = 1, \forall l \in \mathcal{N}_I. \end{aligned} \quad (5)$$

It is obvious that the problem (5) is an intractable non-convex problem because of the presence of the IRS phase shift constraint and the coupling between $\mathbf{P}$ and Φ . However, the traditional solution such as closed-form solution is unique for specific problems and may not be suitable for the considered issue in this manuscript.

IV. PROPOSED SOLUTION BASED ON DRL

To solve problem (5), we need to derive the optimal value of the transmission power at devices $\mathbf{P}$ and IRS phase shift Φ . The selection of these values only depends on the current environment state, having nothing to do with the former action, which can be seen as a Markov decision process. Consequently, DRL emerges as an appropriate solution to our predicament. However, our problem involves a huge and continuous state space, considering the current transmission power and high-dimensional channel state information. Traditional Deep Q Learning with its discrete Q value function is inadequate for resolution. In light of this, we turn to DDPG, which utilizes a Deep Neural Network (DNN) to approximate the value and policy functions, making it fit for handling continuous problems in our problem. Therefore, the minimum secrecy rate in formula (5) is maximized via jointly designing the value of $\mathbf{P}$ and Φ utilizing DDPG. These two variables are seen as action elements and are updated via the trial-and-error interaction between the agent and the environment. We begin with the description of our DDPG structure. Subsequently, we delineate its design. Finally, we give the overall description of our algorithm.

A. Structure of DDPG

Our proposed DDPG is a DRL-based algorithm, which is defined by a four-tuple $\langle A, S, R, P \rangle$.

(1) A : the action space. The agent takes the action $a^t \in A$ in time step t under given state $s^t \in S$ according to the policy $\pi(s^t, a^t)$, and the environment returns a reward r^{t+1} .

(2) S : the state space, representing environmental information. $s^t \in S$ observes the environment in time step t .

(3) R : the reward. r^{t+1} is the reward given by the environment when the agent takes action a^t given s^t . Reward serves as a metric to evaluate the action, and the agent refines the actions based on the reward.

(4) P : the transition function, which denotes the probability from state s^i to state s^j , $i \neq j$.

In DRL, policy, value function, and experience replay are also important parts. The policy $\pi(s^t, a^t)$ is the probability that the agent chooses action a^t under a given state s^t . The value function, similar to the reward, evaluates the quality of an action considering its potential future impact. Similar to Reinforcement Learning (RL), the Q value function $Q(s^t, a^t)$ is adopted as the value function, while it is approximated through a DNN. Then, the Q value function in the DRL can be expressed as:

$$Q(s^t, a^t) \triangleq Q(s^t, a^t | \theta),$$

where θ is the model parameter.

Experience replay mitigates the issue of high correlation in RL, which can impede convergence. It involves a five-tuple $(s^t, a^t, r^{t+1}, s^{t+1}, done)$, where *done* represents the termination state. The system stores these experiences and randomly samples them with a specified size to reduce training sample correlation.

Our proposed DDPG follows an Actor-Critic structure, consisting of two networks: the actor network $\mu(\cdot|\theta^\mu)$ and the critic network $Q(\cdot|\theta^Q)$. θ^μ and θ^Q are the network parameters for updating actor and critic networks, respectively. The actor network is the policy-base network, selecting an action under given s , while critic network is the value-based network, measuring the quality of the chosen action. Receiving the state as input, the actor network generates the corresponding action. Subsequently, the action and the state are taken as the input of the critic network to compute the Q value. To avoid fluctuation and bootstrapping in the training phase, target networks are employed, consisting of target actor and target critic networks denoted as $\mu'(\cdot|\theta^{\mu'})$ and $Q'(\cdot|\theta^{Q'})$, respectively. $\theta^{\mu'}$ and $\theta^{Q'}$ represent the network parameters for updating target actor and target critic network.

In each training iteration, the critic and actor networks are updated. The update of the critic network is to minimize the “Temporal Difference” error, which is the difference between the current Q value and the next-state Q value. The next-state Q value is the target Q value we need to calculate through the target critic network:

$$y = r^{t+1} + \gamma(1 - done)Q'(s^{t+1}, a^{t+1}|\theta^{Q'}), \quad (6)$$

$\gamma \in [0, 1]$ is the discount factor, determining the importance of future rewards [34]. The current Q value is the evaluation value of the current state and action calculated through the critic network:

$$q = Q(s^t, a^t|\theta^Q). \quad (7)$$

Finally, the critic network is updated by performing the Stochastic Gradient Descent (SGD) algorithm on the loss function L_c :

$$\theta^{Q^{t+1}} = \theta^{Q^t} - \mu_c \nabla_{\theta^Q} L_c, \quad (8)$$

where L_c can be given as:

$$L_c = (y - q)^2, \quad (9)$$

$\nabla_{\theta^Q} L_c$ is the gradient of L_c with respect to the critic network parameter θ^Q , and μ_c represents the training rate for updating the critic network.

The update of the actor network is to change the output and the parameters to maximize the cumulative expected reward. It is achieved by calculating the gradient of critic network on the actor output and then increasing the actor output along this gradient to increase the Q value:

$$\theta^{\mu^{t+1}} = \theta^{\mu^t} - \mu_a \nabla_{a^t} Q(s^t, a^t|\theta^Q) \nabla_{\theta^\mu} \mu(s^t|\theta^\mu), \quad (10)$$

where μ_a is the learning rate for updating the actor network, $\nabla_{a^t} Q(s^t, a^t|\theta^Q)$ is the gradient of critic network with respect to a^t , and $\nabla_{\theta^\mu} \mu(s^t|\theta^\mu)$ is the gradient on the actor network with respect to its parameters.

The update on the corresponding target networks is slower than that of primary networks. In our proposed DDPG, a soft update method known as exponential moving average is adopted to update the parameters of target networks. Consequently, the target critic and target actor networks are updated in the following manner:

$$\theta^{Q'} = \tau_c \theta^{Q'} + (1 - \tau_c) \theta^Q, \quad (11)$$

$$\theta^{\mu'} = \tau_a \theta^{\mu'} + (1 - \tau_a) \theta^\mu, \quad (12)$$

where $\tau_c \in (0, 1)$ and $\tau_a \in (0, 1)$ are the learning rate for updating target critic and target actor network, respectively.

B. Construction of the Proposed DDPG

As we mentioned before, there are four DNNs in our proposed DDPG algorithm: actor, critic, target actor, and target critic networks. In our proposed algorithm, both the actor and critic networks share the same structure, comprising an input layer, two hidden layers, and an output layer, as shown in Fig. 2. The target networks are completely the same as their respective primary networks. The actor network accepts the state as input and produces an action as output, hence, the input and output sizes depend on the specific action and state. Similarly, for the critic network, its input dimension relies on the action and state, while its output dimension corresponds to the Q value, since it takes action and state as inputs and yields the Q value as output. The number of hidden layer neurons is up to the number of users, IRS elements, and the BS antennas.

We derive the transmission power $\mathbf{P}$ and IRS phase shift Φ as actions under the given state in the action network. In our approach, these two variables should satisfy the constraints: $|\Phi_l|^2 = 1, l \in \mathcal{N}_I$ and $\mathbf{P} \leq P_{Max}$. To achieve this, we incorporate a normalization step within the actor network, guaranteeing these values do not exceed their own maximum limits. To guarantee the constraint $\mathbf{P} \leq P_{Max}$, we first derive the value of $\mathbf{P}$, then implement the constraint through the following formula:

$$\mathbf{P} = \frac{\mathbf{P}}{\sqrt{P_{Max}}}.$$

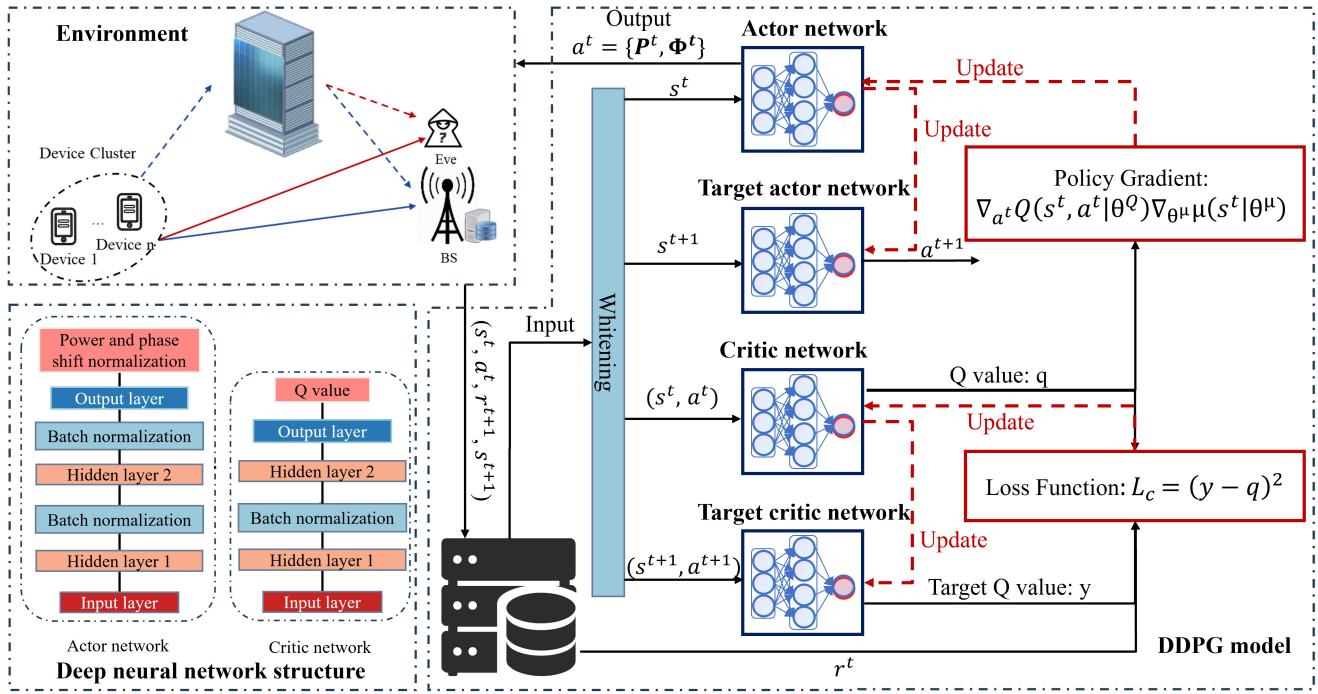


Fig. 2. The structure of proposed DDPG.

Similarly, we normalize the Φ_l to meet the corresponding constraint. The value of Φ_l is a complex number, so its real and imaginary parts should be calculated separately.

$$\Re(\Phi_l) = \frac{\Re(\Phi_l)}{\sqrt{\Re(\Phi_l)^2 + \Im(\Phi_l)^2}},$$

$$\Im(\Phi_l) = \frac{\Im(\Phi_l)}{\sqrt{\Re(\Phi_l)^2 + \Im(\Phi_l)^2}}.$$

The input state for both the actor and critic networks exhibits high correlation characters, which impedes the algorithm convergence rate. To mitigate this issue, we introduce a whitening process prior to feeding the state data into the actor and critic network as shown in Fig. 2, which eliminates the correlation among sample data. Furthermore, to improve training stability and speed, we apply batch normalization before each hidden layer. This reduces the internal covariate shift within the hidden layers' inputs and minimizes the sensitivity of the model to the weight initialization. The $\tanh$ function is employed in every layer to introduce crucial non-linearity to the network, which enables the network to grasp intricate patterns in the input data and facilitate the learning of complex relationships. Moreover, the Adam algorithm serves as the optimizer to leverage a moving average of gradients to expedite convergence. Simultaneously, it incorporates squared gradients to dynamically adjust the learning rates and enhance the adaptability of the optimization process. The overall DDPG structure is presented in Fig. 2.

C. Description of the Algorithm

In our proposed algorithm, the most critical part lies in the interaction between the agent and the environment. The

agent observes the environment to get its state and chooses the action accordingly. In response, the environment gives the reward based on the state and action. Thus, in this subsection, we mainly describe the construction of the state, action, and reward. At the end of this subsection, we describe the proposed algorithm to address the secrecy rate maximization problem.

1) State: the state at time step t is dependent on the current devices' transmission power $\mathbf{P}$, channel information of $\mathbf{h}_{nb}$, $\mathbf{h}_{ne}$, $\mathbf{h}_{ni}$, $\mathbf{H}_{ib}$, and $\mathbf{h}_{ie}$, and the action in the time step $t - 1$. The complex value cannot be directly used as the input of the neural network. Therefore the real and the imaginary part of the complex value should be separated and treated as the independent input of the neural network. The transmission power at user n can be expressed as $p_n = \Re(p_n) + \Im(p_n)$, thus there are $2N_T$ entries contributed by the transmission power. The channel matrices are all complex values, considering every single channel, the entries can be that, $2N_d N_T$ for channel $\mathbf{h}_{nb}$, $2N_d$ for $\mathbf{h}_{ne}$, $2N_d N_I$ for $\mathbf{h}_{ni}$, $2N_I N_T$ for $\mathbf{H}_{ib}$, and $2N_I$ for $\mathbf{h}_{ie}$. The total entries contributed by the channel information is $2(N_T + N_d N_I + N_I N_T + N_d + N_I)$. The size of the action in time step t is $2N_T + 2N_I$. The total dimension of the state space is $D_s = 2(2N_T + N_d + N_d N_T + N_d N_I + N_I N_T + 2N_I)$.

2) Action: in each time step, the action space consists of the transmission power $\mathbf{P}$ and IRS phase shift Φ . Consequently, the dimension of the action space totally depends on the sizes of the above two variables. Moreover, since both these variables are of complex type, the action space dimension is given by $D_a = 2N_T + 2N_I$.

3) Reward: the minimum secrecy rate at users shown in Equation (4) is set as the reward at each time step under the environment state and action derived from the actor network.

The goal of our algorithm is to choose the optimal value of $\mathbf{P}$ and Φ to maximize the minimum secrecy rate. The overall algorithm runs over N episodes, of which each consists of T iterations. At the beginning of the algorithm, the parameters of the actor network θ^μ , critic network θ^Q , target actor network $\theta^{\mu'}$, and target critic network $\theta^{Q'}$, the transmission power matrix $\mathbf{P}$, the IRS phase shift matrix $\text{diag}(\Phi)$, and the experience replay buffer R are initialized. At the beginning of each episode, the state is reset. The agent selects an action using the actor network, which involves computing the values of variables $\mathbf{P}$ and Φ . The next state is observed and the reward can be calculated through Equation (4). These four parameters along with the termination state *done* are assembled and stored into the experience replay buffer R . Subsequently, M samples $(s^i, a^i, r^{i+1}, s^{i+1}, \text{done})$ are randomly chosen from R to update networks. Specifically, the current evaluation Q value is calculated via the critic network, taking (s^i, a^i) as the input. Simultaneously, the current target Q value is derived via the target critic network, utilizing s^{i+1} as the input. Then, the actor and critic networks are updated via minimizing the Equation (9) and maximizing the Equation $\nabla_{a^i} Q(s^i, a^i | \theta^Q)$, respectively. The target network is softly updated using a parameter-wise weighted copy. This is a weighted average of the current target network parameters and the primitive network parameters, where the weight of the current target network parameters is determined by τ .

The algorithm halts when the stopping criterion is met or when the maximum number of iterations is reached. The stopping criterion involves achieving the secrecy rate under ideal conditions, where there is no noise or interference from Eves. The overall algorithm is shown as Algorithm 1.

D. Complexity Analysis

The time complexity of DDPG is primarily determined by the training of the neural network. The computational complexity of neural networks mainly depends on their structure, parameters, and the operations involved in the forward pass, which are determined by the input and output dimensions as well as the number of layers.

The actor and critic networks are fully connected DNNs. The target network is a replica of their primary network, also constituting fully connected DNNs. Our purpose is to derive the optimal actions of transmission power at devices and IRS phase shift, so we aim to train the actor network using critic, target actor, and target critic networks. Thus, the complexity of training the neural network lies in the actor network. As the actor network consists of linear layers, the time complexity of a linear layer can be calculated as $\mathcal{O}(D_s D_a)$, where D_s and D_a are the sizes of input and output, respectively. The additional operations from batch normalization and activation functions contribute to the time complexity, which is typically overshadowed by the linear transformations in the actor network and can be neglected. Consequently, the overall time complexity is roughly the sum of the complexity of each layer and can be given as: $\mathcal{O}(\sum_{l=1}^L D_s^l D_a^l)$, where L denotes the number of layers which

Algorithm 1: DDPG Algorithm for Maximizing the Minimum Secrecy Rate via Joint Design of $\mathbf{P}$ and Φ

Input: $\mathbf{h}_{nb}, \mathbf{h}_{ne}, \mathbf{h}_{ni}, \mathbf{H}_{ib}, \mathbf{h}_{ie}$

Output: Optimal action $a = \{\mathbf{P}, \Phi\}$

Initialize: the actor network parameter θ^μ , the critic network parameter θ^Q , the target actor network parameter $\theta^{\mu'} \leftarrow \theta^Q$, the target critic network $\theta^{Q'} \leftarrow \theta^\mu$, the transmission power $\mathbf{P}$, the IRS phase shift Φ , and the experience replay buffer R .

while True **do**

for $episode \leftarrow 0$ **to** $N - 1$ **do**

 Collecting the channel information to receive the initial observation state s^0 .

for $t \leftarrow 0$ **to** $T - 1$ **do**

 Selecting action $a^t = \mu(s^t | \theta^\mu)$ according to the current policy.

 Executing action a^t and observe the instant reward r^{t+1} , and new state s^{t+1} .

 Storing the experience $(s^t, a^t, r^{t+1}, s^{t+1}, \text{done})$ in R .

 Sampling a random mini-batch of size M from R .

 Computing the target Q value and estimated Q value according to (6) and (7), respectively.

 Constructing training critic network loss function according to (9).

 Executing the SGD on (9) and updating the training critic network according to (8).

 Executing the SGD on

$$q^t = Q(s^t, a^t | \theta^Q)$$

 and update the training actor network according to (10).

 Updating the target networks via (11) and (12).

end

end

end

is four in our current algorithm. D_s^l and D_a^l are the size of input and output in layer l .

V. NUMERICAL RESULTS

In this section, we present the simulation results to evaluate the performance of our proposed algorithm. We assume a two-dimensional Cartesian coordinate system with BS at $(0, 0)m$ and IRS center at $(50, 0)m$. The devices and the Eve are randomly distributed within circles of radius $50m$, centered at $(0, 0)m$ and $(50, 0)m$, respectively. The values of communication and training parameters are specified in Table I. To demonstrate the efficacy of the deployment of IRS and the effectiveness of our algorithm, we compare our algorithm with two benchmark schemes: 1) FL without IRS, and 2) FL with random IRS phase shift.

We first study how the values of P_{Max} and IRS elements affect the performance of DDPG as shown in Figs. 3 and 4.

TABLE I
VALUES OF KEY PARAMETERS

Description	Parameter	Value
Maximum transmit power at devices	P_{max}	30 dBW
Reference distance	d_0	1 m
Path loss at reference distance	PL_0	-30 dB
Learning rate for updating actor network	μ_a	0.001
Learning rate for updating critic network	μ_c	0.001
Learning rate for updating target actor network	τ_a	0.001
Learning rate for updating target critic network	τ_c	0.001
Discounted rate for future reward	γ	0.99
Buffer size	D	100000
Batch size	M	16
Maximum number of episodes	N	5000
Number of steps in each episode	T	10000

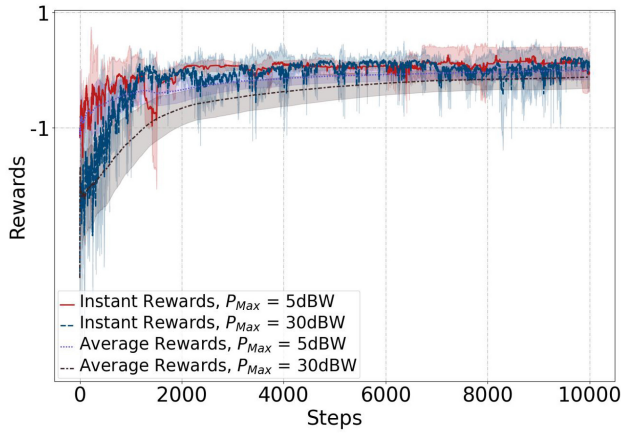


Fig. 3. Training rewards with different P_{Max} values.

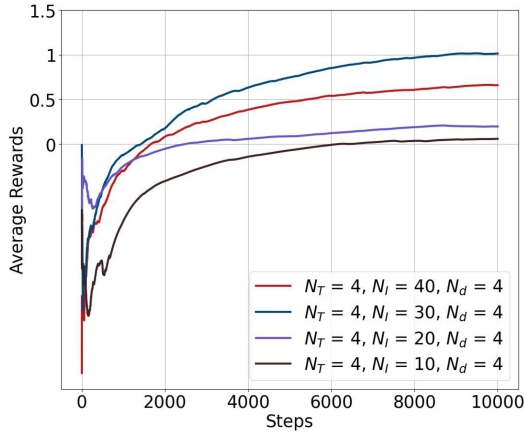


Fig. 4. Average rewards with different numbers of IRS.

Notably, the convergence is finally reached under two specific scenarios. In Fig. 3, we unveil the effect of different P_{Max} values on the algorithm rewards including instant rewards and average rewards. The average reward is the cumulative expected reward defined as the average value from the beginning to the current position. As shown in Fig. 3, a lower P_{Max} value leads to faster convergence of the algorithm for the reason that a higher transmission power increases the interference and degrades the model update. At the same

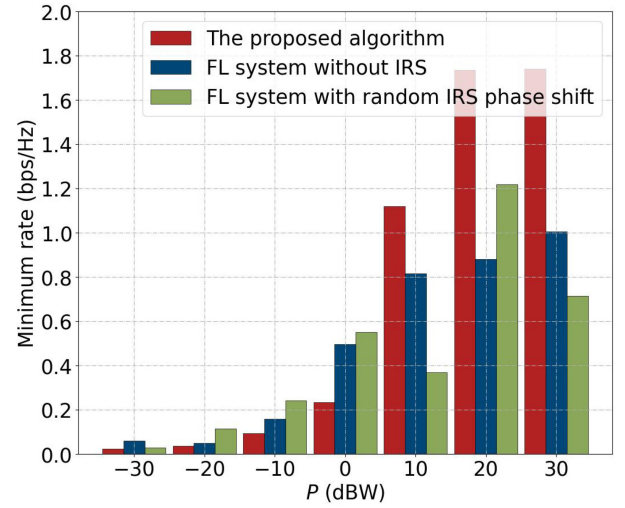


Fig. 5. The minimum secrecy rate with varying transmission power.

time, a lower transmission power results in a higher channel quality by reducing the noise impact. We further investigate the impact of IRS element number N_I on the DDPG algorithm performance. From Fig. 4, we observe that increasing N_I degrades the algorithm's convergence, while increasing N_I can generally improve the average reward when the value of N_I does not reach 30. When the value is beyond 30, the average rewards start to decline due to excessive computational overhead.

Then, we examine the impact of the variables on the minimum secrecy rate. An increased secrecy rate can be observed as the transmission power grows in Fig. 5, which can be attributed to that a higher transmission power at devices enhances the communication link between the device and BS. The deployment of IRS further amplifies this signal by intelligently reflecting and focusing the signal towards the BS, simultaneously greatly mitigating the signal to Eve. Notably, that enhancement becomes more pronounced with higher transmission power. Moreover, our proposed algorithm demonstrates that the minimum secrecy rate increases as the IRS element number rises, outperforming the scheme without IRS as illustrated in Fig. 6. This can be attributed to that the increased N_I enables the system to focus signals with greater precision, directing them toward specific receivers. Furthermore, we observe that the scheme utilizing random IRS phase shift exhibits lower effectiveness compared to our proposed algorithm. It also exhibits fluctuations with an increase in the number of IRS elements, indicating that only the IRS with a well-designed phase shift can work effectively.

We further investigate the effect of the transmission power under two different system settings, namely, $N_T = 4$, $N_d = 4$, $N_I = 4$, $E = 1$ and $N_T = 10$, $N_d = 10$, $N_I = 4$, $E = 1$ as shown in Fig. 7. As we mentioned before, the minimum secrecy rate increases with respect to the transmission power at the device. Additionally, we can observe from Fig. 7, under the same value of transmission power P , the increasing number of devices results in the reduced minimum secrecy rate. This

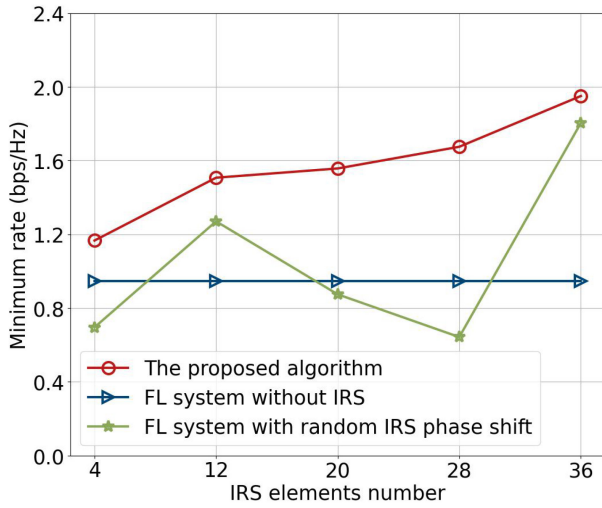


Fig. 6. The minimum secrecy rate with different numbers of IRS elements.

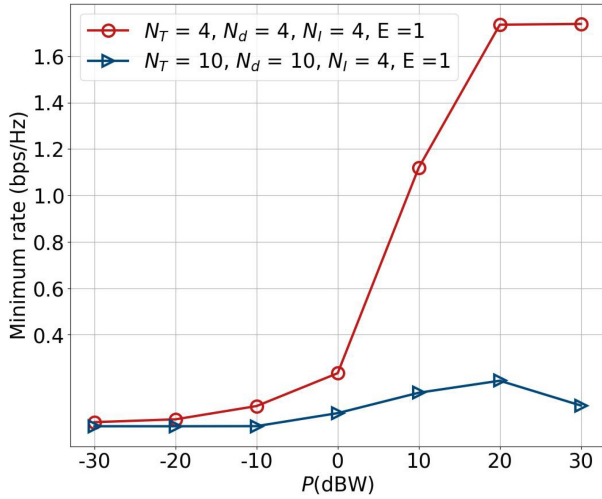


Fig. 7. The minimum secrecy rate with varying transmission power in two considered scenarios.

is because that the increasing number of devices accessing the network results in the competition of limited resources such as frequency bands and time slots, which further leads to congestion and decreased performance. On the other hand, adding more antennas to the BS may not always increase the quality of communication channels. There are diminishing returns as the number of antennas increases. To a certain extent, the benefits of additional antennas may be outweighed by factors like signal processing complexity and interference.

Finally, we study how variations in the path loss exponent of the IRS-Device channel impact the minimum secrecy rate. We can observe from Fig. 8 that when $N_d = N_T = 4$, the IRS-Device path loss has a significant impact on the secrecy rate, and increasing the path loss can enhance the secrecy rate. Moreover, the impact of path loss on the secrecy rate varies with the number of BS antennas and users since the BS antenna configuration and user density can significantly change the impact of path loss on the secrecy rate in MU-MIMO. The effect of path loss on secrecy rate is comparatively smaller with massive antennas and users.

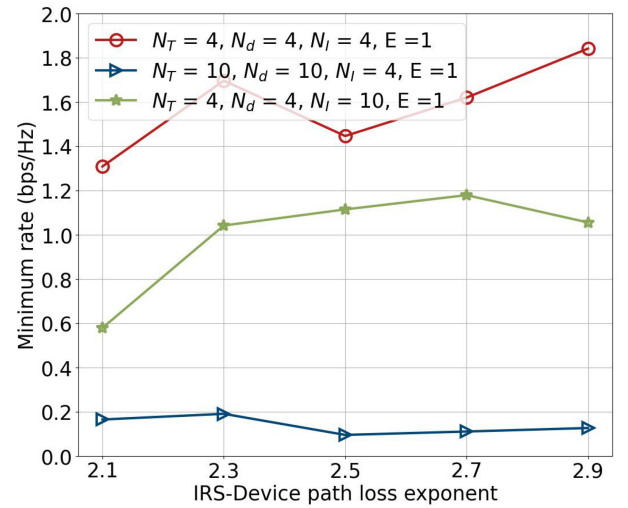


Fig. 8. The minimum secrecy rate with varying IRS-Device path loss in two considered scenarios.

VI. CONCLUSION

In this article, we investigate the IRS-assisted NOMA-based FL system, where Eves can overhear the model parameters sent from devices to BS. To mitigate the influence of unsatisfied wireless channels and secure the FL training, we formulate a minimum secrecy rate maximization problem where the devices' transmission power and IRS phase shift should be jointly designed. To address this problem, the DRL-based algorithm with the DDPG model is presented. Simulation results demonstrate that the secrecy rate can be significantly improved via deploying the IRS with a well-designed phase shift, which implies that the IRS can effectively enhance the legitimate channel while degrading the wire-tap channel.

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